1. *Contextual documentation* - a visual record of the physical state at a point in time, useful for tracking degradation or construction progress.
2. *Source of structured data* computer vision can automatically extract information like equipment IDs, valve positions, gauge readings, and corrosion levels, digitizing operational state data.
3. *Metadata* - contains useful information like location, position, and camera pose.

TRY IT OUT: EXTRACT STRUCTURED DATA FROM A PHOTOGRAPH

In the context of building my home digital twin, photographs are an important source of data related to utility consumption. Since utility bills are only generated quarterly, I do not have real time consumption data to inform decisions and actions. I can take photographs of utility meters and then use computer vision to extract data from the analog water and gas meters into a machine readable format and update the state within the digital twin of my home at a much higher frequency than digitizing quarterly bills. This is demonstrated in figure 2.3 using a photo of my water meter taken with a smartphone, and the same library, Tesseract, as I used to extract text from the PDF bill.



Figure 2.3 A photograph of my residential water meter (1), with the area of interest extracted and converted to grayscale in (2) before being inverted in (3) to enable OCR extraction of the digits.

The code, shown in listing 2.2, is slightly more complex due to the need to crop the area in the image where the meter digits are and preprocess the image to get an optimal result. Figure 2.3 shows the progression of the image through the extraction process. This code will extract the number